Competitive Exam Practice 4: Expression Notation, Recursion, and Tower of Hanoi

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. Descriptive = write the missing code/text.

Q1 *[GATE CS, verified text — year uncertain: GATEOverflow V2 p.231 heads it 1997 Q1.7; ExamSIDE lists it under 2004]* [**MCQ, 1 mark**] Which of the following is essential for converting an infix expression to the postfix form efficiently?

(A) An operator stack

(B) An operand stack

(C) An operand stack and an operator stack

(D) A parse tree

Q2 *[GATE CSE 2004 Q38, verified — GATEOverflow V2 p.231]* [**MCQ, 2 marks**] Assume that the operators `+`, `-`, `*` are left-associative and `^` is right-associative. The order of precedence (from highest to lowest) is `^`, `*`, `+`, `-`. (Below, `*` stands for ×.) The postfix expression corresponding to the infix expression `a+b*c-d^e^f` is

(A) `abc*+def^^-`

(B) `abc*+de^f^-`

(C) `ab+c*d-e^f^`

(D) `-+a*bc^^def`

Q3 *[GATE CSE 2007 Q38, verified — GATEOverflow V2 p.232]* [**MCQ, 2 marks**] The following postfix expression with single digit operands is evaluated using a stack:

```
8 2 3 ^ / 2 3 * + 5 1 * -
```

Note that `^` is the exponentiation operator. The top two elements of the stack after the first `*` is evaluated are:

(A) `6, 1`

(B) `5, 7`

(C) `3, 2`

(D) `1, 5`

Q4 *[GATE CSE 2002 (descriptive), verified text / single-source year — GATEOverflow V2 p.310]* [**Descriptive, 2 marks**] The following recursive function in C is a solution to the Towers of Hanoi problem. Fill in the dotted parts of the solution.

```
void move(int n, char A, char B, char C) {
    if (......................) {
        move (............................);
        printf("Move disk %d from pole %c to pole %c\n", n, A, C);
        move (.....................);
    }
}
```

Q5 *[Original, GATE-pattern]* [**MCQ, 1 mark**] The prefix expression corresponding to the infix expression `(a+b)*(c-d)` (usual precedence; `+` and `-` left-associative) is:

(A) `*+ab-cd`

(B) `*-ab+cd`

(C) `++*ab-cd`

(D) `*+ab+cd`

Q6 *[Original, GATE-pattern]* [**MCQ, 1 mark**] The postfix expression corresponding to the prefix expression `+*ab-cd` is:

(A) `ab+cd-*`

(B) `ab*cd-+`

(C) `ab*cd+-`

(D) `cd-ab*+`

Q7 *[Original, GATE-pattern]* [**NAT, 1 mark**] The Tower of Hanoi problem with 6 disks is solved by the standard recursive algorithm (move $n-1$ disks out of the way, move the largest disk, restack the $n-1$ disks). Counting one disk move as one step, the total number of moves is ____________. (Answer in integer)

Q8 *[Original, GATE-pattern]* [**MCQ, 2 marks**] For the Tower of Hanoi problem with $n = 6$ disks solved by the standard recursive algorithm, let M be the total number of disk moves and D be the maximum number of simultaneously live activation frames of the recursive procedure. Then (M, D) equals:

(A) $(64, 6)$

(B) $(63, 6)$

(C) $(63, 63)$

(D) $(31, 5)$